

BENSON, United Kingdom

WMO: 036580

Lat: 51.6203N

Lon: 1.0986W

Elev: 187

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	25.3	19.0	14.4	25.2	22.0	16.7	28.3	28.9	48.2	26.2	49.3	2.4	50	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	81.4	65.0	77.4	63.4	74.0	62.1	66.9	77.1	65.1	73.8	63.6	71.3	9.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	87.5	70.0	61.8	83.1	68.2	60.4	79.1	66.8	31.4	76.8	30.1	73.9	29.0	71.3	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.9	18.7	DB	15.2	87.1	3.8	3.9	12.5	89.9	10.3	92.2	8.2	94.4	5.4	97.3	(4)
(5)			WB	14.6	70.2	3.8	1.9	11.9	71.6	9.7	72.7	7.6	73.8	4.8	75.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	39.8	40.4	43.4	47.5	52.9	58.4	62.6	62.0	57.2	51.5	44.7	40.8		
	DBStd	9.95	6.52	6.22	5.61	5.55	5.47	4.82	4.67	4.72	4.76	5.81	6.40	7.36		
	HDD50	1472	317	273	213	109	32	1	0	0	3	51	179	294		
	HDD65	5482	780	690	669	524	375	206	104	119	237	418	609	751		
	CDD50	1532	2	2	9	35	122	253	392	371	220	98	20	8		
	CDD65	69	0	0	0	0	1	8	31	25	4	0	0	0		
	CDH74	772	0	0	0	3	16	116	365	236	33	3	0	0		
	CDH80	163	0	0	0	0	0	22	85	55	1	0	0	0		
(14) Wind	WSAvg	8.5	9.7	9.6	9.2	8.6	8.5	7.9	7.6	7.5	7.7	8.7	8.4	9.1		
(15) Precipitation	PrecAvg	26.6	2.6	1.9	1.6	2.1	2.1	1.9	1.9	2.2	2.1	2.7	2.9	2.5		
	PrecMax	34.6	6.1	4.5	3.5	5.4	5.2	5.4	4.4	5.1	4.4	5.4	5.7	4.1		
	PrecMin	19.4	0.5	0.2	0.4	0.1	0.2	0.4	0.4	0.2	0.4	0.9	1.1	0.5		
	PrecStd	4.6	1.3	1.1	0.8	1.3	1.1	1.1	1.0	1.3	1.2	1.3	1.3	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	56.7	62.5	71.5	76.2	83.1	86.7	85.6	78.3	69.2	60.3	57.3		
		MCWB	52.6	51.9	52.2	57.3	61.7	65.8	66.3	66.7	63.9	61.9	57.0	54.1		
	2%	DB	53.6	53.6	58.2	65.2	71.9	77.2	82.0	80.4	73.2	64.6	58.1	55.3		
		MCWB	51.0	49.9	50.6	54.7	60.6	63.3	64.8	65.3	62.2	58.5	55.1	52.8		
	5%	DB	51.6	51.6	55.2	61.1	67.8	72.8	78.0	75.7	69.4	62.4	56.3	53.7		
		MCWB	48.8	48.3	49.4	52.3	58.5	61.5	63.5	63.2	60.0	57.6	53.4	51.3		
	10%	DB	49.7	49.8	52.8	57.7	63.6	69.3	74.1	71.9	66.5	60.4	54.4	51.8		
		MCWB	47.2	46.8	48.2	50.5	55.7	59.7	62.2	61.5	59.0	56.5	51.7	49.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	53.2	53.9	59.5	64.9	68.3	68.8	69.3	66.3	62.7	57.9	54.8		
		MCDB	54.7	56.1	58.5	68.7	72.3	80.1	81.4	81.2	73.9	67.8	59.7	56.2		
	2%	WB	51.4	50.6	52.0	55.8	61.7	64.7	66.6	66.6	63.8	60.2	55.7	53.4		
		MCDB	53.5	52.9	56.0	63.2	69.5	73.3	77.6	75.9	70.4	63.3	57.6	55.1		
	5%	WB	49.3	48.9	50.6	53.5	59.2	62.7	65.0	64.7	62.0	58.7	53.8	51.7		
		MCDB	51.3	51.2	54.4	59.4	66.2	70.6	74.7	72.2	67.3	61.3	55.8	53.6		
	10%	WB	47.6	47.3	49.0	51.5	57.0	60.9	63.4	63.0	60.3	57.2	52.0	49.6		
		MCDB	49.5	49.6	52.4	56.7	62.9	68.2	71.9	69.6	64.9	60.0	54.1	51.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.3	15.0	17.6	17.7	18.4	19.3	18.4	17.4	14.5	12.2	11.0		
		MCDBR	11.6	12.6	18.4	23.1	24.1	25.3	26.7	24.4	21.7	15.0	11.1	9.9		
		MCWBR	10.8	10.2	12.3	13.8	14.1	13.5	11.8	11.1	12.0	10.2	9.4	9.4		
	5% WB	MCDBR	11.6	12.0	15.9	20.7	21.0	21.7	22.6	20.1	18.2	13.1	10.9	10.0		
		MCWBR	11.1	10.4	11.4	12.9	12.9	12.7	11.4	10.7	11.4	10.0	10.0	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.342	0.378	0.401	0.405	0.403	0.399	0.389	0.372	0.355	0.329	0.316		
	taud		2.359	2.369	2.265	2.215	2.242	2.275	2.296	2.344	2.389	2.417	2.406	2.344		
	Ebn at Noon		214	246	256	264	269	270	269	266	257	237	212	197		
	Edh at Noon		20	26	34	40	41	40	39	35	30	24	19	17		
(44) All-Sky Solar Radiation	RadAvg		257	465	807	1241	1508	1663	1577	1310	988	575	306	203		
	RadStd		21	61	109	145	96	170	159	106	85	51	35	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (85 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page